

Annex 176D

(normative)

Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M

176D.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-module 200 Gb/s one-lane Attachment Unit Interface (200GAUI-1 C2M), 400 Gb/s two-lane Attachment Unit Interface (400GAUI-2 C2M), 800 Gb/s four-lane Attachment Unit Interface (800GAUI-4 C2M), and 1.6 Tb/s eight-lane Attachment Unit Interface (1.6TAUI-8 C2M). These interfaces have specified electrical characteristics at accessible compliance points, which may be used for building and testing systems with pluggable modules.

The 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M are physical instantiations of the PMA service interface, as described in 176.2, which enable partitioning a Physical Layer implementation across multiple devices. Within this annex, these interfaces are collectively referenced as 200 Gb/s per lane AUI-C2M.

Figure 176D–1 shows the relationship of the 200 Gb/s per lane AUI-C2M, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. Physical Layer partitioning options using 200 Gb/s per lane AUI-C2M are provided in Annex 176B.

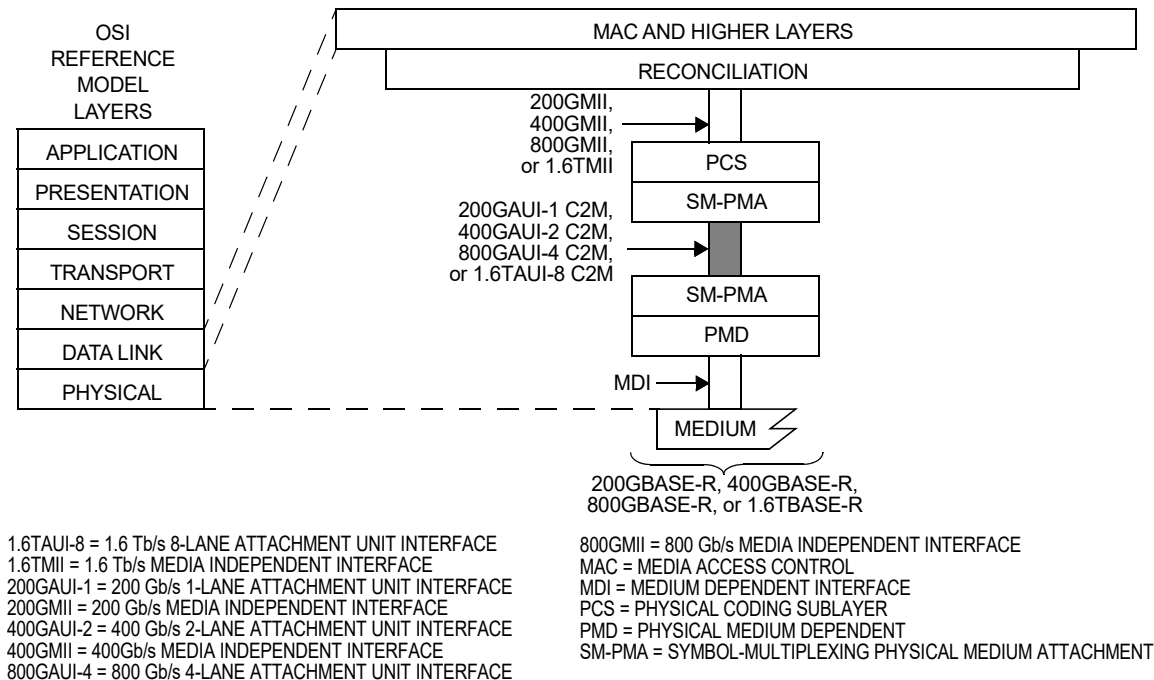


Figure 176D–1—200 Gb/s per lane AUI-C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

176D.2 Error ratio allocation

With a compliant input signal, a C2M component receiver (see 176D.6.6 and 176D.6.7) is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA adjacent to the C2M component using the method described in 174A.9, with BER_{added} equal to 2.681×10^{-4} .

176D.3 Functional specification

A 200 Gb/s per lane AUI-C2M is a bidirectional data path. Each direction of a 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M or 1.6TAUI-8 C2M data path contains one, two, four, or eight lanes, respectively.

A 200 Gb/s per lane AUI-C2M is specified in terms of a host and a module. The host and the module both contain components that implement the interface, referred to as C2M components.

An n -lane C2M component is functionally equivalent to a corresponding n -lane PMD specified in Clause 179. The C2M component shall meet the functional specifications in 179.8 unless stated otherwise.

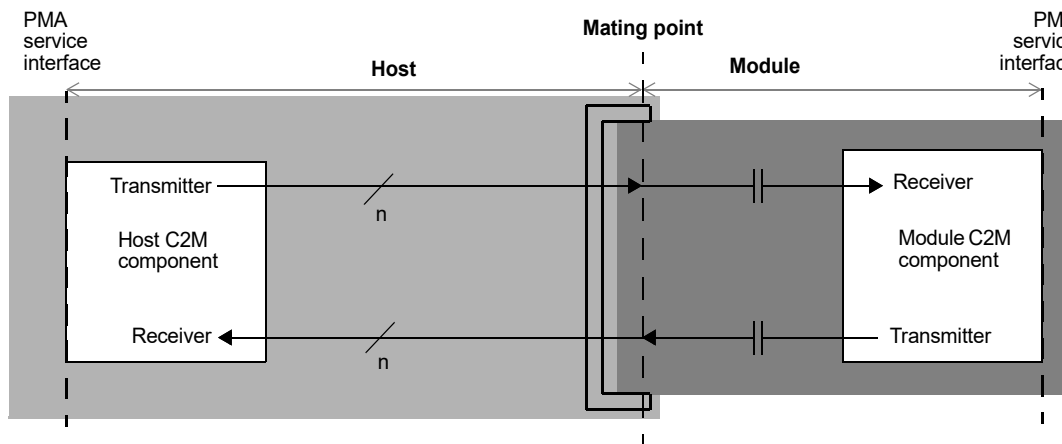
NOTE 1—As part of the functional equivalence to a PMD, C2M components include the inter-sublayer link training (ILT) function with E1 format and the ready-to-send (RTS) function, as specified in Annex 178B.

Each lane consists of one differential transmission line in each direction between the host C2M component and the module C2M component. The transmission lines are AC-coupled within the module and have a common ground reference.

The electrical characteristics of the host are specified in 176D.6.4 and 176D.6.6. The electrical characteristics of the module are specified in 176D.6.5 and 176D.6.7.

NOTE 2—The host and module electrical specifications are not equivalent to those of the corresponding PMDs defined in Clause 179. Specifically, the test points at which module compliance is defined are different.

Figure 176D–2 depicts a typical 200 Gb/s per lane AUI-C2M application. The reference characteristics of the channel between the C2M components are provided in 176D.7.



NOTE 1—The number of lanes n is 1 for 200GAUI-1, 2 for 400GAUI-2, 4 for 800 GAUI-4, and 8 for 1.6TAUI-8.
NOTE 2—The host is depicted as having a receptacle into which the module is plugged. This is based on the MDI connectors specified in Annex 179C. Other connection methods may be used.

Figure 176D-2—200 Gb/s per lane AUI-C2M link diagram

176D.4 Service interfaces

The PMA above the 200 Gb/s per lane AUI-C2M is any $m:1$ PMA for 200GAUI-1, $m:2$ PMA for 400GAUI-2, $m:4$ PMA for 800GAUI-4, and $m:8$ PMA for 1.6TAUI-8, as specified in Clause 176. The PMA below the 200 Gb/s per lane AUI-C2M is any $1:n$ PMA for 200GAUI-1, $2:n$ PMA for 400GAUI-2, $4:n$ PMA for 800GAUI-4, and $8:n$ PMA for 1.6TAUI-8, as specified in Clause 176.

The service interface above and below the 200 Gb/s per lane AUI-C2M is the PMA service interface as specified in 176.2.

For an AUI component above the AUI channel, the SIGNAL_OK parameter of the PMA:IS_SIGNAL.indication primitive corresponds to the variable rts_status of the RTS function, as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of the PMA:IS_DATAUNIT_ i .indication primitive on all lanes are unspecified.

For an AUI component below the AUI channel, the SIGNAL_OK parameter of the PMA:IS_SIGNAL.request primitive corresponds to the variable rts_status of the RTS function, as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the tx_symbol parameters of the PMA:IS_DATAUNIT_ i .request primitive on all lanes are unspecified.

The service interfaces are illustrated in Figure 176D–3.

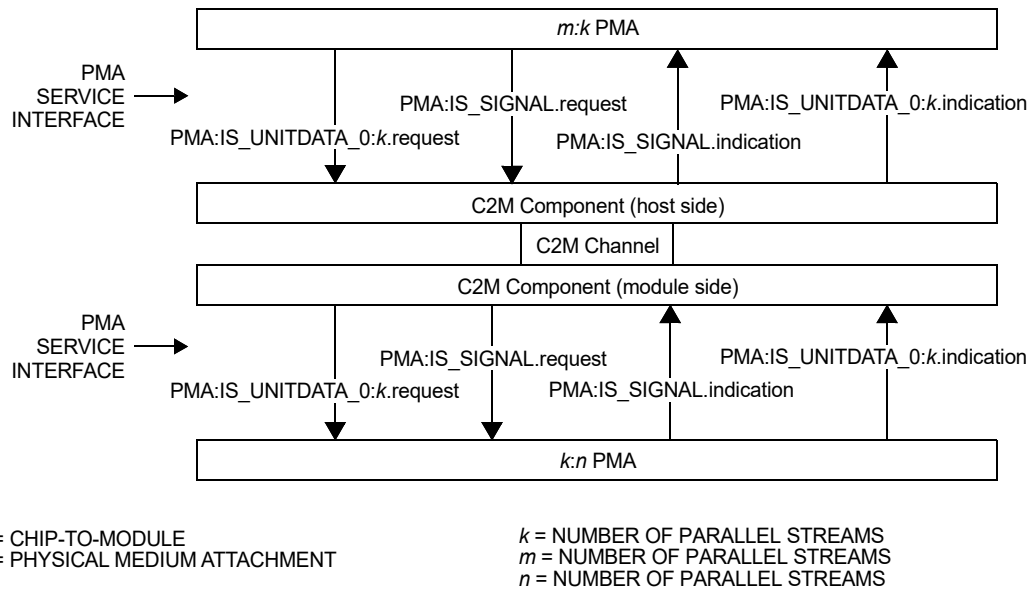


Figure 176D–3—Inter-sublayer service interfaces for a 200GAUI-1, 400GAUI-2, 800GAUI-4, or 1.6TAUI-8 C2M

176D.5 Delay constraints

The sum of the transmit and receive delays contributed by a 200 Gb/s per lane AUI-C2M component including the AUI-C2M channel in one direction shall be no more than the maximum value listed in Table 176D–1. It is assumed that the one-way delay through the channel is no more than 5 ns.

Table 176D–1—Delay constraints

Component	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GAUI-1	9 216	18	46.08
400GAUI-2	18 432	36	46.08
800GAUI-4	36 864	72	46.08
1.6TAUI-8	73 728	144	46.08

^a One bit time is equal to 5 ps for 200GAUI-1, 2.5 ps for 400GAUI-2, 1.25 ps for 800GAUI-4, and 625 fs for 1.6TAUI-8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GAUI-1, 1.28 ns for 400GAUI-2, 640 ps for 800GAUI-4, and 320 ps for 1.6TAUI-8. See 1.4.459 for the definition of pause_quanta.

Descriptions of overall system delay constraints can be found in 116.4 for 200GAUI-1 and 400GAUI-2, in 169.4 for 800GAUI-4, and in 174.4 for 1.6TAUI-8.

176D.6 Electrical characteristics

176D.6.1 Compliance points

The electrical characteristics for the C2M host and module are defined at specified compliance points. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters. A Host Compliance Board (HCB) is mechanically equivalent to a module and can be plugged into a host. A Module Compliance Board (MCB) is mechanically equivalent to a host and enables a module to be plugged in.

Figure 176D–4 depicts the location of compliance points in which host characteristics are specified. The test points are separate for each lane. The output of the HCB at TP1a is used to verify the host electrical output compliance. The input of the HCB at TP4a is used to verify the host input compliance.

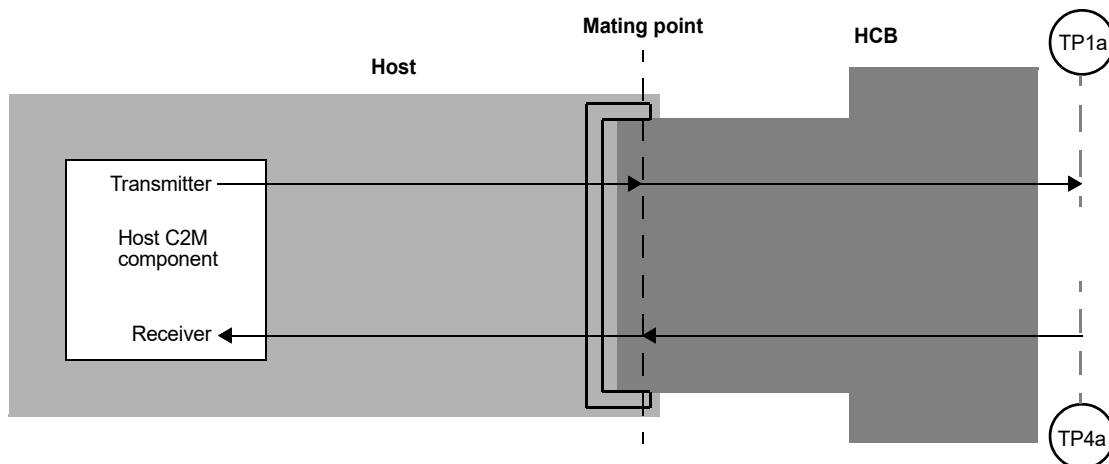


Figure 176D–4—Host compliance points

NOTE 1—TP1a and TP4a correspond to the Clause 179 test points TP2 and TP3, respectively.

Figure 176D–5 depicts the location of compliance points in which module characteristics are specified. The test points are separate for each lane. The output of the MCB at TP4 is used to verify the module electrical output compliance. The input of the MCB at TP1 is used to verify the module input compliance.

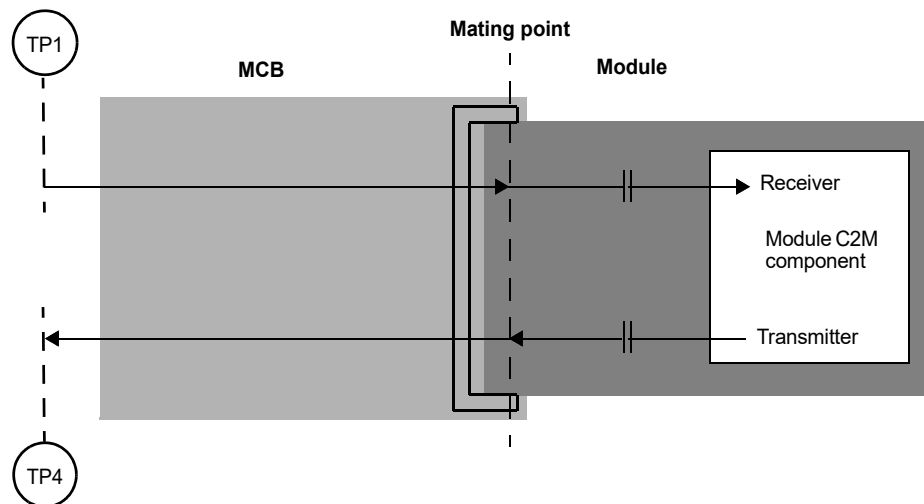


Figure 176D–5—Module compliance points

NOTE 2—The test points defined above are not necessarily equivalent to the outputs of the channel or the inputs of the host and module C2M components.

176D.6.2 HCB/MCB characteristics

The HCB characteristics are described in 179B.2, where the HCB performs the equivalent function as the TP2 or TP3 test fixture.

The MCB characteristics are described in 179B.3, where the MCB performs the equivalent functionality as the cable assembly test fixture.

The characteristics of the compliance boards (HCB and MCB) in mated state are described in 179B.4, where the mated compliance boards are equivalent to the mated test fixtures (MTF).

The AUI-C2M connector mentioned in this annex is equivalent to the MDI connector in 179B.2 and 179B.3.

176D.6.3 Reference impedance

The reference impedance for differential specifications is $92.5\ \Omega$. The reference impedance for common-mode specifications is $23.125\ \Omega$.

NOTE—Calculation of modal S-parameters may require transformation of the measured single-ended S-parameters to the specified reference impedance. See 178A.1.3.

Unless specified otherwise, specified parameters are measured on a load or termination equal to the reference impedance.

176D.6.4 Host output characteristics

A host shall meet the host output specifications given in Table 176D–2 on each lane.

Unless specified otherwise, host output signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP1a to $50\ \Omega$ single-ended loads in the test equipment.

Table 176D–2—Summary of host output specifications at TP1a

Parameter	Reference	Value	Units
Signaling rate (range)	176D.8.1	106.25 ± 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.2	0.03	V
Output disabled		1	V
Output enabled			
DC common-mode voltage (range)	176D.8.2	0.2 to 1	V
AC common-mode peak-to-peak voltage (max)	176D.8.2	0.032	V
Low-frequency, $V_{CM_{LF}}$		0.085	V
Full-band, $V_{CM_{FB}}$			
Effective return loss, ERL (min)	176D.8.3.1	7.3	dB
Common-mode to common-mode return loss	176D.8.3.4	Table 176D–9	—
Common-mode to differential-mode return loss	176D.8.3.4	Table 176D–9	—

Table 176D–2—Summary of host output specifications at TP1a (continued)

Parameter	Reference	Value	Units
Transmitter steady-state voltage, v_f (range)	176D.8.4	0.372 to 0.5	V
Linear fit pulse peak ratio, R_{peak} (min)	176D.8.4	0.161	—
Level separation mismatch ratio, R_{LM} (min)	176D.8.5	0.95	—
Transmitter output equalization	176D.8.6	0.005 to 0.025	—
absolute value of step size for all taps (range)			
value at minimum state for $c(-3)$ (max)			
value at maximum state for $c(-2)$ (min)			
value at minimum state for $c(-1)$ (max)			
value at minimum state for $c(0)$ (max)			
value at minimum state for $c(1)$ (max)			
Signal-to-noise-and-distortion ratio, $SNDR$ (min)	176D.8.7	Table 179–9	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176D.8.8	26	dB
Output jitter (max)	176D.8.9	0.023	UI
JH _{RMS}			
EOJ ₀₃			
JH4u			

176D.6.5 Module output characteristics

A module shall meet the module output specifications given in Table 176D–3.

The module output shall be AC-coupled within the module, with low-frequency 3 dB cutoff frequency less than 250 kHz.

Unless specified otherwise, module output signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with DC-coupled connection from TP4 to 50 Ω single-ended loads in the test equipment.

Table 176D–3—Summary of module output specifications at TP4

Parameter	Reference	Value	Units
Signaling rate (range) ^a	176D.8.1	106.25 $\pm$ 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.2	0.03	V
Output disabled		1	V
Output enabled			
DC common-mode voltage tolerance (range)	176D.8.2	0.15 to 1.05	V
AC common-mode peak-to-peak voltage (max)	176D.8.2	0.015	V
Low-frequency, $V_{CM_{LF}}$		0.06	V
Full-band, $V_{CM_{FB}}$			
Effective return loss, ERL (min)	176D.8.3.1	8.5	dB
Common-mode to common-mode return loss	176D.8.3.4	Table 176D–9	—
Common-mode to differential-mode return loss	176D.8.3.4	Table 176D–9	—
Transmitter steady-state voltage, v_f (range)	176D.8.4	0.392 to 0.5	V

Table 176D–3—Summary of module output specifications at TP4 (continued)

Parameter	Reference	Value	Units
Linear fit pulse peak ratio, R_{peak} (min)	176D.8.4	0.43	—
Level separation mismatch ratio R_{LM} (min)	176D.8.5	0.95	—
Transmitter output equalization absolute value of step size for all taps (range) value at minimum state for $c(-3)$ (max) value at maximum state for $c(-2)$ (min) value at minimum state for $c(-1)$ (max) value at minimum state for $c(0)$ (max) value at minimum state for $c(1)$ (max)	176D.8.6	0.005 to 0.025 –0.06 0.12 –0.34 0.5 –0.2	— — — — — —
Signal-to-noise-and-distortion ratio, $SNDR$ (min)	176D.8.7	Table 179–9	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176D.8.8	28	dB
Output jitter (max) JH_{RMS} EOJ_{03} JH_{4u}	176D.8.9	0.023 0.025 0.118	UI UI UI

^a The signaling rate is derived from the PMD receiver input.

176D.6.6 Host input characteristics

A host shall meet the host input specifications given in Table 176D–4.

Table 176D–4—Summary of host input specifications at TP4a

Parameter	Reference	Value	Units
Signaling rate (range)	176D.8.1	106.25 ± 50 ppm	GBd
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, VCM_{LF} Full-band, VCM_{FB}	176D.8.10	0.015 0.06	V V
Effective return loss, ERL (min)	176D.8.3.1	7.3	dB
Differential-mode to common-mode return loss	176D.8.3.4	Table 176D–9	—
Transmitter output amplitude tolerance	176D.8.14	0.5	V
Interference tolerance	176D.8.12	Table 176D–12	—
Jitter tolerance	176D.8.13	Table 176D–13	—
DC common-mode voltage (range) ^a	176D.8.2	0.2 to 1	V

^a Generated by host, referred to host ground.

176D.6.7 Module input characteristics

The module input shall be AC-coupled within the module. It is recommended that the low-frequency 3 dB cutoff frequency be less than 250 kHz.

The module input shall meet the specifications given in Table 176D–5.

Table 176D–5—Summary of module input specifications at TP1

Parameter	Reference	Value	Units
Signaling rate (range)	176D.8.1	106.25 ± 50 ppm	GBd
DC common-mode voltage tolerance (range)	176D.8.10	0.15 to 1.05	V
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	176D.8.10	0.032 0.085	V V
Differential-mode to common-mode return loss	176D.8.3.4	Table 176D–9	—
Effective return loss, ERL (min)	176D.8.3.1	8.5	dB
Transmitter output amplitude tolerance	176D.8.14	0.5	V
Interference tolerance	176D.8.12	Table 176D–12	—
Jitter tolerance	176D.8.13	Table 176D–13	—

176D.7 Expected channel properties

Unlike the related C2C interface specified in Annex 176C, the channel between the C2M components is not specified, since it is divided between two entities with different compliance requirements. The C2M components in the host and in the module with their respective portions of the channel are specified by their input and output characteristics at the compliance points specified in 176D.6.1.

This subclause describes the expected properties of the channels from TP0d to TP1d and from TP4d to TP5d, as depicted in Figure 176D–6. TP0d and TP5d represent the host C2M component output and input, respectively, at the device-to-package interface, similar to the definitions of these test points in Table 179–6. TP1d and TP4d represent the module C2M component input and output, respectively, at the device-to-package interface. These test points are typically not accessible in an implemented system.

The content of this subclause is a reference model that may be used for host and module design. It is expected that the normative input and output specifications of host and module in this annex can be met with a variety of implementation approaches.

176D.7.1 Reference insertion loss budget

Figure 176D–6 depicts the reference differential insertion loss (IL_{dd}) values at 53.125 GHz for specific parts of the channel between the C2M components. The insertion losses of the host channel, the module channel, and the TP0d-TP1d and TP4d-TP5d channels are not expected to be measurable.

176D.7.2 COM reference model

The 200 Gb/s per lane AUI-C2M channel is associated with a channel operating margin (COM) reference model, which defines the assumed capabilities of the transmitter and receiver functions of the C2M components.

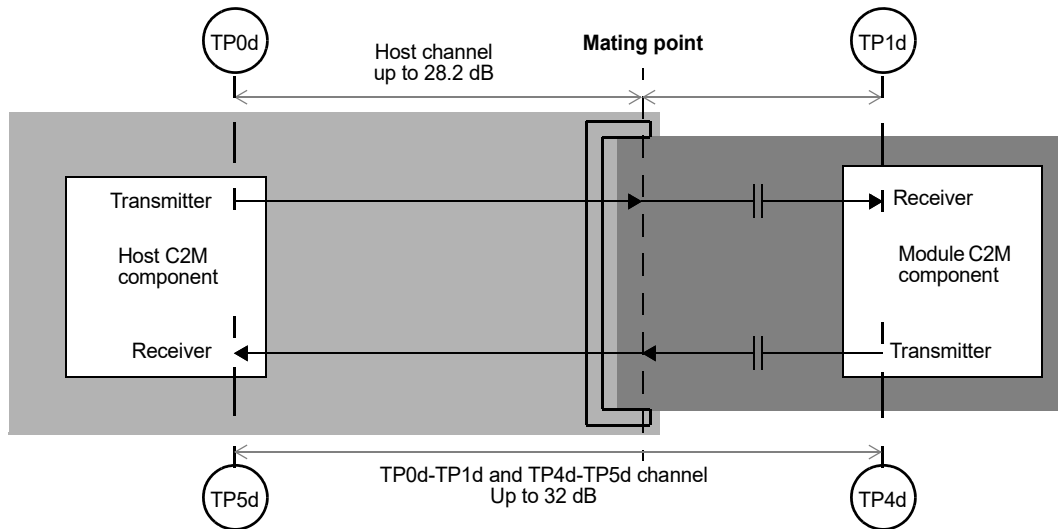


Figure 176D-6—Reference insertion loss budget at 53.125 GHz

COM calculation, as defined in 178A.1, is used for calibration of noise in the interference tolerance test (see 176D.8.12). Calculation of COM is performed with the parameters in this subclause.

The host parameters in Table 176D-6 represent a host channel that includes a combination of a packaged device and a PCB, using the S-parameter model defined in 178A.1.4 and 178A.1.5. This model is used for host input test calibration and is not a host channel specification.

The parameters in Table 176D-7 correspond to behavioral models of transmitters and receivers that, with suitable host and module channel models, are compliant to the host and module specifications in this clause. The purpose of these parameters is to compute COM, and they do not represent requirements for transmitter and receiver implementations.

It is expected that the specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation and host channel construction.

NOTE—COM is not a specification for hosts and modules. COM analysis (see Annex 178A) may be used informatively for host design.

Table 176D-6—Host and module model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF

Table 176D–6—Host and module model parameters (*continued*)

Parameter	Symbol	Value	Units
Device package model, host			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	45	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Partial host channel model			
Single-ended board capacitance at the package-to-board interface	C_0	0	nF
Transmission line parameter γ_0	$\gamma_0^{(h)}$	0	1/mm
Transmission line parameter a_1	$a_1^{(h)}$	5.95×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	$a_2^{(h)}$	2.6×10^{-5}	ns/mm
Transmission line parameter τ	$\tau^{(h)}$	5.79×10^{-3}	ns/mm
Transmission line characteristic impedance	$Z_c^{(h)}$	92.5	Ω
Transmission line length	$z_p^{(h)}$	280	mm
Single-ended capacitance at the model-to-measurement interface	C_1	0	nF
Single-ended reference resistance	R_0	46.25	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE 1—The device package and partial host channel model parameters are based on the reference impedance R_0 . NOTE 2—The host device package model is similar to the “class B” device package model defined in Table 176C–9 but with a specific value for “Transmission line 1 length”.			

Table 176D–7—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r		
Host to module direction		0.004	ns
Module to host direction		0.007	ns

Table 176D–7—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$		
Minimum value		0	
Maximum value		0	—
Step size		0	
Transmitter equalizer, coefficient –2	$c(-2)$		
Minimum value		0	
Maximum value		0.14	—
Step size		0.02	
Transmitter equalizer, coefficient –1	$c(-1)$		
Minimum value		–0.34	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, coefficient 0	$c(0)$		—
Minimum value		0.54	
Transmitter equalizer, coefficient 1	$c(1)$		
Minimum value		–0.2	
Maximum value		0	—
Step size		0.02	
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
Random jitter, RMS	σ_{RJ}	0.01	UI
One-sided noise spectral density	η_0	7.5×10^{-9}	V ² /GHz
Noise-equivalent quantization bits	N_{qb}	6	—
Quantization clipping probability	P_{qc}	10^{-7}	

Table 176D-7—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	5	—
Number of fixed-position taps	N_{fix}	14 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	56 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	−0.7 −0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-5}	—
Include maximum likelihood sequence detection		no	—

^aThere are 8 fixed-position post-cursor taps.

^bThe latest post-cursor position for a floating tap is 50.

176D.8 Methodology

176D.8.1 Signaling rate

A host or module shall comply with all output requirements in 176D.8.4 through 176D.8.9 for any output signaling rate in the specified range.

A host or module shall comply with all input tolerance requirements in 176D.8.12 through 176D.8.14 for any input signaling rate in the specified range.

NOTE—For the module output, and for the host output when the Physical Layer includes an AUI-C2C interface, the signaling rate is derived from the signaling rate of a signal received from another transmitter.

176D.8.2 Maximum voltages

Differential and common-mode signals are defined in 93.8.1.3.

Peak-to-peak output voltages are defined to a probability P , with respect to the distribution of the output voltage V_{out} sampled at an effective rate of at least two samples per UI. The sampling may be either synchronous or asynchronous.

A maximum output voltage is defined as the value V_{max} such that the probability that $V_{out} > V_{max}$ is $P/2$. A minimum output voltage is defined as the value V_{min} such that the probability that $V_{out} < V_{min}$ is $P/2$. A peak-to-peak output voltage is defined as $V_{max} - V_{min}$.

Differential peak-to-peak output voltage is defined with $P=10^{-7}$ for the differential output signal. For compliance testing, it is sufficient to measure it with the lane under test transmitting scrambled idle, while lanes not under test transmit either scrambled idle or PRBS31Q.

Full-band AC common-mode peak-to-peak voltage, $VC_{M_{FB}}$, is defined with $P=10^{-7}$ for the common-mode output signal. For compliance testing, it is sufficient to measure it from the PRBS13Q test pattern, while lanes not under test transmit PRBS31Q.

Low-frequency AC common-mode peak-to-peak voltage, $VC_{M_{LF}}$, is defined with $P=10^{-7}$ for the common-mode output signal filtered by a low-pass filter as defined by Equation (93A–20) with f_r set to 100 MHz. For compliance testing, it is sufficient to measure it from the PRBS13Q test pattern, while lanes not under test transmit PRBS31Q.

NOTE 1—AC common-mode measurement may be sensitive to mismatches between the single-ended paths in the test fixture and the test setup. Careful design and calibration of the test system are recommended.

NOTE 2—AC common-mode measurement should take into consideration frequencies down to the specified AC-coupling frequency.

176D.8.3 Return loss specifications

176D.8.3.1 Effective return loss (ERL)

ERL is computed using the procedure in 93A.5 with the values in Table 176D–7 and Table 176D–8, with differential reference impedance equal to $92.5\ \Omega$, and with T_{fx} equal to twice the test fixture delay. T_{fx} is provided by the test fixture provider.

For host input and output ERL, the test fixture delay is defined as the propagation delay between the coaxial connector on the test fixture and the host-facing connection, excluding the discontinuity at the HCB-to-host mating interface. For module input and output ERL, the test fixture delay is defined as the propagation delay between the coaxial connector on the test fixture and the first discontinuity of the AUI-C2M connector.

Table 176D–8—ERL parameter values

Parameter	Symbol	Value		Units
		Host input and output	Module input and output	
Transition time associated with a pulse	T_r	0.005		ns
Incremental available signal loss factor	β_x	0		GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618		—
Length of the reflection signal	N	1600	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	0	UI
Tukey window flag	tw	1		—
Number of samples per unit interval	M	32		—

176D.8.3.2 Modal effective return loss

Modal effective return loss (ERL_{CC} , ERL_{CD} , or ERL_{DC}) is the effective return loss computed according to 93A.5 using the values in Table 176D–7 and Table 176D–8 and T_{fx} as defined in 176D.8.3.1, with the following exceptions.

- For host output ERL_{CC} , $s_{ii}(f)$ (see 93A.5.1) is replaced with the host output $s_{CC22}(f)$.
- For host output ERL_{DC} , $s_{ii}(f)$ is replaced with the host output $s_{DC22}(f)$.
- For host input ERL_{CD} , $s_{ii}(f)$ is replaced with the host input $s_{CD11}(f)$.
- For module output ERL_{CC} , $s_{ii}(f)$ is replaced with the module output $s_{CC22}(f)$.
- For module output ERL_{DC} , $s_{ii}(f)$ is replaced with the module output $s_{DC22}(f)$.
- For module input ERL_{CD} , $s_{ii}(f)$ is replaced with the module input $s_{CD11}(f)$.

$s_{CC22}(f)$, $s_{DC22}(f)$, and $s_{CD11}(f)$ are defined in 176D.8.3.3.

176D.8.3.3 Mixed-mode reflection coefficients

Mixed-mode reflection coefficients can be computed from two-port S-parameter measurements at transmitter outputs or receiver inputs. The two transmitter output ports are labeled 3 and 4 and the two receiver input ports are labeled 1 and 2.

The differential reference impedance is 92.5 Ω and the common-mode reference impedance is 23.125 Ω . This corresponds to a single-ended reference impedance of 46.25 Ω . The reference impedance for measured S-parameters may be transformed to the required reference impedance using the procedure defined in 178A.1.3.

The transmitter output common-mode to common-mode reflection coefficient is defined by Equation (176D–1).

$$s_{CC22}(f) = (s_{33}(f) + s_{34}(f) + s_{43}(f) + s_{44}(f)) / 2 \quad (176D-1)$$

The transmitter output common-mode to differential-mode reflection coefficient is defined by Equation (176D–2).

$$s_{DC22}(f) = (s_{33}(f) + s_{34}(f) - s_{43}(f) - s_{44}(f)) / 2 \quad (176D-2)$$

The receiver input differential-mode to common-mode reflection coefficient is defined by Equation (176D–3).

$$s_{CD11}(f) = (s_{11}(f) - s_{12}(f) + s_{21}(f) - s_{22}(f)) / 2 \quad (176D-3)$$

176D.8.3.4 Modal return loss specifications

The common-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179–14) or having ERL_{CC} value higher than the specification in Table 176D–9.

The common-mode to differential-mode return loss specification is satisfied by either meeting the requirement of Equation (179–15) or having ERL_{DC} higher than the value in Table 176D–9.

The differential-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179–22) or having ERL_{CD} higher than the value in Table 176D–9.

Table 176D–9—Modal return loss specifications

Specification	RL mask equation	RL mask figure	Modal ERL type	Modal ERL minimum (dB)
Common-mode to common-mode	Equation (179–14)	Figure 179–4	ERL _{CC}	2
Common-mode to differential-mode	Equation (179–15)	Figure 179–5	ERL _{DC}	17
Differential-mode to common-mode	Equation (179–22)	Figure 179–7	ERL _{CD}	17

176D.8.4 Steady-state voltage and linear fit pulse peak ratio

Steady-state voltage (v_f) and linear fit pulse peak ratio (R_{peak}) are defined in 179.9.4.2.2.

176D.8.5 Level separation mismatch ratio

Level separation mismatch ratio (R_{LM}) is defined in 120D.3.1.2.

Host and module output are required to meet the R_{LM} (min) requirement in Table 176D–2 and Table 176D–3, respectively, when the transmit equalization is set to each of the initial conditions defined in Table 179–8.

176D.8.6 Transmit equalization

Host and module outputs include programmable transmit equalization as specified in 179.9.4.2 that is controlled by the ILT function with a training frame of E1 format, as specified in Annex 178B. Host output step size and coefficient limits are specified in Table 176D–2. Module output step size and coefficient limits are specified in Table 176D–3.

176D.8.7 Signal-to-noise-and-distortion ratio

Signal-to-noise-and-distortion ratio ($SNDR$) for module output is defined at TP4 and calculated using the method in 179.9.4.6.

$SNDR$ for host output is defined at TP1a and calculated using the method in 179.9.4.6, with the exception that the host inputs at TP4a on each lane are driven by asynchronous signals with transmit equalization (see 176D.8.6) set to PRESET 1, with steady-state voltage at the maximum value specified in Table 176D–2 and transition time of 6 ps. The asynchronous signal pattern is either PRBS31Q or PCS data.

NOTE—The input signals are included to excite the NEXT that will be present when a module is plugged into the host under test.

The minimum $SNDR$ requirements in Table 179–9 apply for both module output and host output.

176D.8.8 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio (SNR_{ISI}) is defined in 179.9.4.4.

176D.8.9 Output jitter

The output jitter parameters JH_{RMS} , JH_{4u} , and EOJ_{03} are defined in 179.9.4.7.

176D.8.10 Common-mode voltage tolerance

DC common-mode voltage (see 93.8.1.3) is generated by the host. A compliant module meets the module input and output specifications with any DC common-mode voltage, measured at TP4 and TP1, within the ranges specified in Table 176D–3 and Table 176D–5.

Maximum peak-to-peak AC common-mode voltage is specified using the parameters VCM_{LF} and VCM_{FB} (see 176D.8.2) for both host output and module output. A compliant host meets the host input specifications with both VCM_{LF} and VCM_{FB} , measured at TP4, up to the limits specified in Table 176D–4. A compliant module meets the module input specifications with both VCM_{LF} and VCM_{FB} , measured at TP1a, up to the limits specified in Table 176D–5.

NOTE—The specified voltages allow for the effects of ground offset voltage.

176D.8.11 Error ratio requirements for receiver tests

A receiver on each lane is required to meet the expected BLER specified in 176D.2, using the test method in either 174A.9.5, 174A.9.7, or 174A.9.6 with parameters provided in Table 176D–10. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 176D–11.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of the slope of $\log_{10}(H_m(k))$ for $k=0$ to n , where n is at least 4 and $H_m(k) = 0$ for all $k \geq n$. For the linear fit, $H_m(n)$ is set to $3/N$ where N is the total number of test blocks, and the intercept $H_m(0)$ is set to 1. The fit is performed only for values of k that have $H_m(k) > 0$.

Table 176D–10—Error ratio parameters

Parameter name	Value
p	
200GAUI-1	1
400GAUI-2	2
800GAUI-4	4
1.6TAUI-8	8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	2.681×10^{-4}
BER_{max}	2.4×10^{-5}

Table 176D–11—Receiver error mask

Test symbol errors per test block, k (see 174A.9.5)	$H_{\max}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
1	1.1×10^{-1}	6.1×10^{-2}	3.2×10^{-2}	1.6×10^{-2}
2	7.5×10^{-3}	2×10^{-3}	5.1×10^{-4}	1.3×10^{-4}
3	3.2×10^{-4}	4.3×10^{-5}	5.5×10^{-6}	6.8×10^{-7}
4	1.1×10^{-5}	6.9×10^{-7}	4.4×10^{-8}	2.7×10^{-9}
5	2.7×10^{-7}	8.9×10^{-9}	2.8×10^{-10}	8.2×10^{-12}
6	5.9×10^{-9}	9.5×10^{-11}	1.5×10^{-12}	2.1×10^{-14}
7	1.1×10^{-10}	8.7×10^{-13}	6.5×10^{-15}	4.4×10^{-17}
8	1.7×10^{-12}	6.9×10^{-15}	2.5×10^{-17}	8×10^{-20}
9	2.5×10^{-14}	4.9×10^{-17}	8.6×10^{-20}	1.3×10^{-22}
10	3.2×10^{-16}	3.1×10^{-19}	2.6×10^{-22}	1.8×10^{-25}
11	3.7×10^{-18}	1.8×10^{-21}	7.2×10^{-25}	2.3×10^{-28}
12	4×10^{-20}	9.2×10^{-24}	1.8×10^{-27}	2.6×10^{-31}
13	3.9×10^{-22}	4.4×10^{-26}	4.1×10^{-30}	2.7×10^{-34}
14	3.6×10^{-24}	2×10^{-28}	8.7×10^{-33}	2.6×10^{-37}
15	3×10^{-26}	8.1×10^{-31}	1.7×10^{-35}	2.2×10^{-40}
16	2.4×10^{-28}	3.1×10^{-33}	3.1×10^{-38}	1.8×10^{-43}

176D.8.12 Interference tolerance

Interference tolerance is defined by the procedure described in 176D.8.12.1 through 176D.8.12.4.

176D.8.12.1 Test setup

Host interference tolerance is defined based on the test setup shown in Figure 176D–7a. The test channel is shown in Figure 176D–7b. The test is performed with addition of noise at the pattern generator, calibrated

such that the channel operating margin (COM) calculated with the test channel and the transmitter's measured parameters matches the value in Table 176D–12.

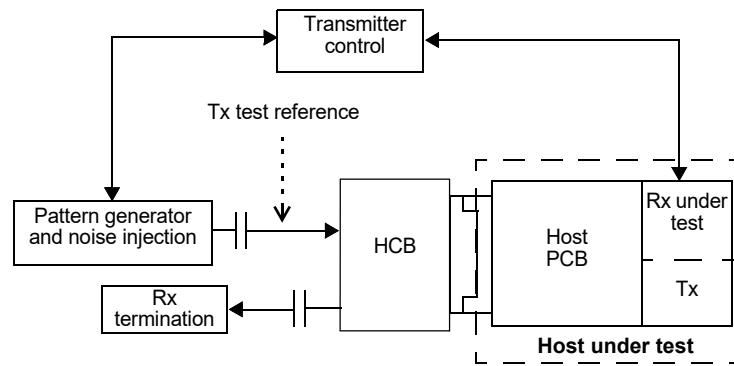


Figure 176D–7a—Host interference tolerance test setup

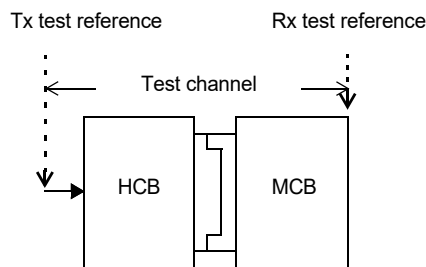


Figure 176D–7b—Host test channel calibration

Module interference tolerance is defined based on the test setup shown in Figure 176D–8a. The test channel is shown in Figure 176D–8b. Two tests are defined, corresponding to low-loss and high-loss host channel.

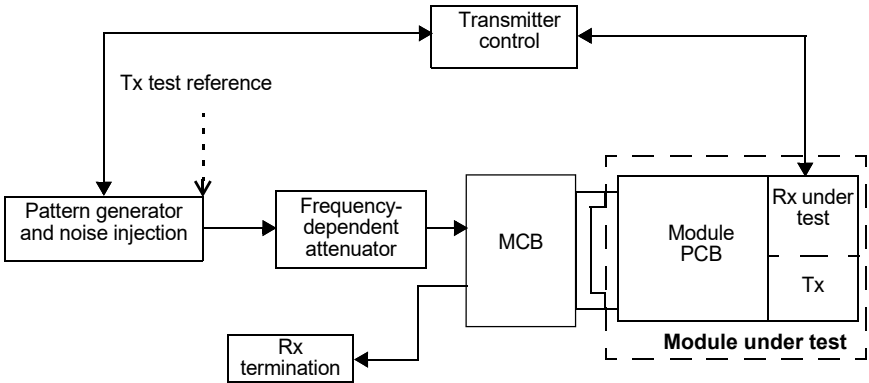


Figure 176D–8a—Module interference tolerance test setup

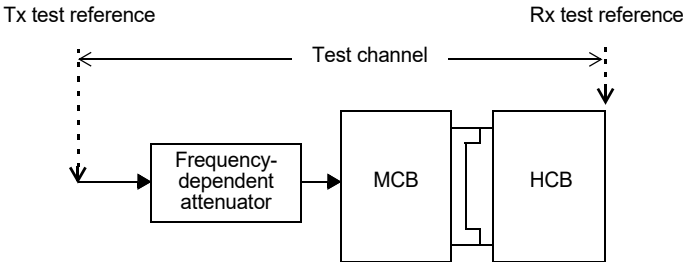


Figure 176D–8b—Module test channel calibration

The test parameters are specified in Table 176D–12.

Table 176D–12—Interference tolerance test parameters

Parameter	Host test	Module test L (low loss) ^a	Module test H (high loss)	Units
Test channel insertion loss at 53.125 GHz	8 ± 2	8 ± 2	32 ± 0.5	dB
Host channel parameters	Table 176D–6	N/A	N/A	
Test pattern	PRBS31Q			
COM ^b	3			dB

^a The low loss test channel consists of mated MCB and HCB with no frequency-dependent attenuator.

^b COM is calculated as defined in 176D.8.12.2. Meeting the test requirements with a lower value of COM demonstrates margin to the specification but is not required for compliance.

176D.8.12.2 Test channel calibration

The COM of the test channel is calculated using the method defined in Annex 178A and the parameters of 176D.7.2, with the following considerations:

- a) For the host test, the test channel is measured between the Tx and Rx test references shown in Figure 176D–7b, and COM is calculated using the partial host channel, device package, and device termination models in Table 176D–6 for the receiver S-parameter model.
- b) For the module test, the test channel is measured between the Tx and Rx test references shown in Figure 176D–8b, and COM is calculated using the device termination models in Table 176D–6 for the receiver S-parameter model.
- c) The transition time T_r is measured at the Tx test reference with the transmit equalizer set to PRESET 1 (i.e., equalization off with maximum amplitude), using the method in 120E.3.1.5, with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz.
- d) EO_{J03} , JH4u, and JH_{RMS} are measured at the Tx test reference with pattern generator noise injection disabled (see 176D.8.12.3). For host test, the jitter complies with the module output specification in Table 176D–3. For module test, the jitter complies with the host output specification in Table 176D–2. In the calculation of COM, A_{DD} and σ_{RJ} are calculated from the measured values of JH4u and JH_{RMS} using Equation (179–16) and Equation (179–17), replacing the values in Table 176D–7. It is recommended to adjust the pattern generator jitter such that JH4u and JH_{RMS} are as close as practical to their limits.
- e) In the calculation of the channel S-parameters in 178A.1.2, the transmitter S-parameter model $S^{(t)}$ is omitted. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A–13) uses a rise time filter $H_t(f)$ (see 178A.1.6.2) calculated with T_r measured in step c.
- f) In the calculation of COM, SNR_{TX} is set to the SNDR value with $N_p=55$ measured at the Tx test reference with pattern generator noise disabled.
- g) The interference noise (see 178A.1.7.5) is derived from the S-parameters of the test channel. For a given RMS noise voltage σ_{ns} measured at the Tx test reference, the spectral density of the noise source $N_s(f)$ used for calculation of COM may be calculated using Equation (179–18). Alternatively, $N_s(f)$ may be obtained by measurement of the noise spectral density at the output of the noise source.
- h) The value of σ_{ns} that yields the target value of COM specified in Table 176D–12 is determined (e.g., via iterative calculation of COM). The noise level at the Tx test reference is then configured so that the measured RMS voltage matches σ_{ns} .

176D.8.12.3 Pattern generator and noise injection

The pattern generator transmits data to the device under test. At the start of transmitter training, the pattern generator output amplitude is set such that steady-state voltage v_f (see 176D.8.4) at the Tx test reference is the minimum specified in Table 176D–3 for host test, or in Table 176D–2 for module test. v_f does not exceed that minimum during the test. The pattern generator equalization is set to the INITIALIZE setting in Table 179–8. During the test, the pattern generator equalization does not exceed the minimum coefficient range in 179.9.4.2.5.

Broadband noise is added to the signal before the Tx test reference, with noise level set according to step h) in 176D.8.12.2.

176D.8.12.4 Test procedure

The pattern generator is first configured to transmit training frames as defined in 178B.7.3.1 with the injected noise as specified in 176D.8.12.3. During this initialization period, the device under test (DUT) configures the pattern generator transmit equalizer coefficients and precoding to the settings it would select

using the ILT function (see 176D.8.6). The settings may be communicated via the ILT function or by other means.

After the pattern generator has been configured and the receiver reports that it is ready (using the signal_ok variable), the pattern generator is set to generate a PRBS31Q pattern. During the test, transmitters in all lanes of the AUI component under test transmit either scrambled idle or PRBS31Q, with equalization turned off (PRESET 1 condition).

A host or module shall meet the error ratio requirements defined in 176D.8.11 under the conditions of this test.

176D.8.13 Jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 176D.8.13.1 and 176D.8.13.2.

176D.8.13.1 Test setup

For hosts, jitter tolerance is measured with the setup described by Figure 176D–7a. The pattern generator includes sinusoidal jitter injection and complies with the module output jitter specifications in Table 176D–3.

For modules, jitter tolerance is measured with a channel described by Figure 176D–8a that meets the insertion loss of Module test H as specified in Table 176D–12. The pattern generator includes sinusoidal jitter injection and complies with the host output jitter specifications in Table 176D–2.

176D.8.13.2 Test procedure

The jitter tolerance test procedure is similar to that of 176D.8.12, with the following exceptions:

- For each case in Table 176D–13, jitter with the specified frequency is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency at the Tx test reference (see Figure 176D–7a and Figure 176D–8a).
- The test channel calibration (see 176D.8.12.2) is performed with the transmitter output set to the jitter frequency and amplitude for Case G of Table 176D–13. The additive noise level obtained is used for all other cases.

A host or module shall meet the error ratio requirements defined in 176D.8.11 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 176D–13.

Table 176D–13—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Case G	Units
Jitter frequency	0.04	0.1333	0.4	1.333	4	12	40	MHz
Jitter amplitude (peak-to-peak)	5	1.5	0.5	0.15	0.05	0.05	0.05	UI

NOTE—For a module input test, A_{DD} and σ_{RJ} calculated from pattern generator measurements using Equation (179–16) and Equation (179–17) can be higher than the values in Table 176D–7. In this case, a suitable channel should be chosen in order to meet the COM requirement with these higher values. For a host input test, if the calculated COM is lower than the requirement, a better pattern generator is needed.

176D.8.14 Transmitter output amplitude tolerance

Transmitter output amplitude tolerance of a receiver is defined as the maximum transmitter steady-state voltage that the receiver can tolerate, such that it satisfies the error ratio requirements of 176D.8.11 when the transmit function operates in DATA mode (see 179.8.2).

The transmitter steady-state voltage is measured as specified in 176D.8.4 at the output of the pattern generator used in the test. The differential peak-to-peak output voltage of the pattern generator does not exceed the maximum value specified in Table 176D–2 (for module testing) or in Table 176D–3 (for host testing).

The pattern generator is initially configured to transmit training frames as defined in 178B.7.3 with transmit equalization at the INITIALIZE setting (see Table 179–8). During the initialization period, the device under test (DUT) may further configure the pattern generator transmit equalizer and precoding settings it would select using the ILT function (see 176D.8.6). The settings may be communicated via the ILT function or by other means. After the pattern generator has been configured, the BLER is measured as specified in 174A.9 or 174A.11.

For a host, the input signal is applied at TP4a and measured at TP4. For a module, the input signal is applied at TP1 and measured at TP1a.

NOTE—Steady-state voltage is defined with PRESET 1. It is not initially generated by a transmitter, due to the INITIALIZE setting in Table 179–8. The receiver is not required to tolerate PRESET 1 unless it specifically requests it.

176D.9 Management variables

AUI component control and status variables intended to be accessible via a management system are listed in Table 179–26 and Table 179–27. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

NOTE—Although the variable names and descriptions in Table 179–26 and Table 179–27 include the term “PMD”, the same variables names are used in AUI components.

If the MDIO Interface (see Clause 45) is implemented, AUI component control and status variables shall be mapped to MDIO registers as shown in Table 179–26, Table 179–27, Table 178B–3, and Table 178B–4, with the following exceptions:

- For a host-side AUI component (see Figure 176D–3), which is the lower AUI component (see 178B.3), the MMD number is different.
- For a module-side AUI component (see Figure 176D–3), which is the upper AUI component (see 178B.3), all register numbers are offset by 4000 (see 45.2.1.273).

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

176D.10 Protocol implementation conformance statement (PICS) proforma for Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M⁷

176D.10.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

176D.10.2 Identification

176D.10.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

176D.10.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

176D.10.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
AUI1	200GAUI-1 C2M	176D.1	One data path in each direction	O/1	Yes [] No []
AUI2	400GAUI-2 C2M	176D.1	Two data paths in each direction	O/1	Yes [] No []
AUI4	800GAUI-4 C2M	176D.1	Four data paths in each direction	O/1	Yes [] No []
AUI8	1.6TAUI-8 C2M	176D.1	Eight data paths in each direction	O/1	Yes [] No []
*HOST	AUI-C2M host	176D.3	AUI component is in a host	O/2	Yes [] No []
*MODL	AUI-C2M Module	176D.3	AUI component is in a module	O/2	Yes [] No []
*MD	MDIO capability	176D.9	Clause 45 MDIO is implemented with the management variable mapping in 179.14	O	Yes [] No []

176D.10.4 PICS proforma tables for 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M

176D.10.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	Functionally equivalent to a PMD	176D.3	Functionally equivalent to a PMD specified in Clause 179 (see 179.8)	M	Yes []
FS2	Host service interface	176D.4	Interface above is the PMA service interface of 176.2	HOST:M	Yes [] N/A []
FS3	Module service interface	176D.4	Interface below is the PMA service interface of 176.2	MODL:M	Yes [] N/A []
FS4	Delay constraints	176D.5	Sum of transmit and receive delays is within constraints	M	Yes []
FS5	Management variables	176D.9		M	Yes []

176D.10.4.2 Electrical specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Host output specifications	176D.6.4	Complies with all specifications in Table 176D–2	HOST:M	Yes []
ES2	Module output specifications	176D.6.5	Complies with all specifications in Table 176D–3	MODL:M	Yes []
ES3	Host input specifications	176D.6.6	Complies with all specifications in Table 176D–4	HOST:M	Yes []
ES4	Module input specifications	176D.6.7	Complies with all specifications in Table 176D–5	MODL:M	Yes []

176D.10.4.3 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	176D.9		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	176D.9		MD:M	Yes [] N/A []